

Bluestock Mutual Fund Analytics Capstone Project

Executive Summary

This project presents an end-to-end Mutual Fund Analytics solution developed using Python, SQLite, Jupyter Notebooks, and Power BI. The objective was to build a financial intelligence platform capable of processing mutual fund data, evaluating fund performance, analyzing investor behavior, and visualizing key insights through interactive dashboards.

Project Overview

The Mutual Fund Analytics project covers the complete analytics lifecycle:

- Data Ingestion
- Data Cleaning & Transformation
- Database Management
- Exploratory Data Analysis
- Performance Analytics
- Advanced Risk Analytics
- Dashboard Development

The project uses multiple mutual fund datasets including NAV history, scheme performance, investor transactions, benchmark indices, SIP inflows, portfolio holdings, and fund master information.

Data Engineering Pipeline

A Python-based ETL pipeline was developed to automate data processing.

ETL Workflow

1. Raw data ingestion
2. Data validation and cleaning
3. Feature engineering
4. Processed CSV generation
5. SQLite database loading

Scripts Developed

- data_ingestion.py
- clean_nav_history.py
- clean_transactions.py
- clean_performance.py
- clean_remaining_datasets.py
- load_to_sqlite.py

- etl_pipeline.py

The ETL pipeline successfully executes all processing steps and loads cleaned datasets into SQLite.

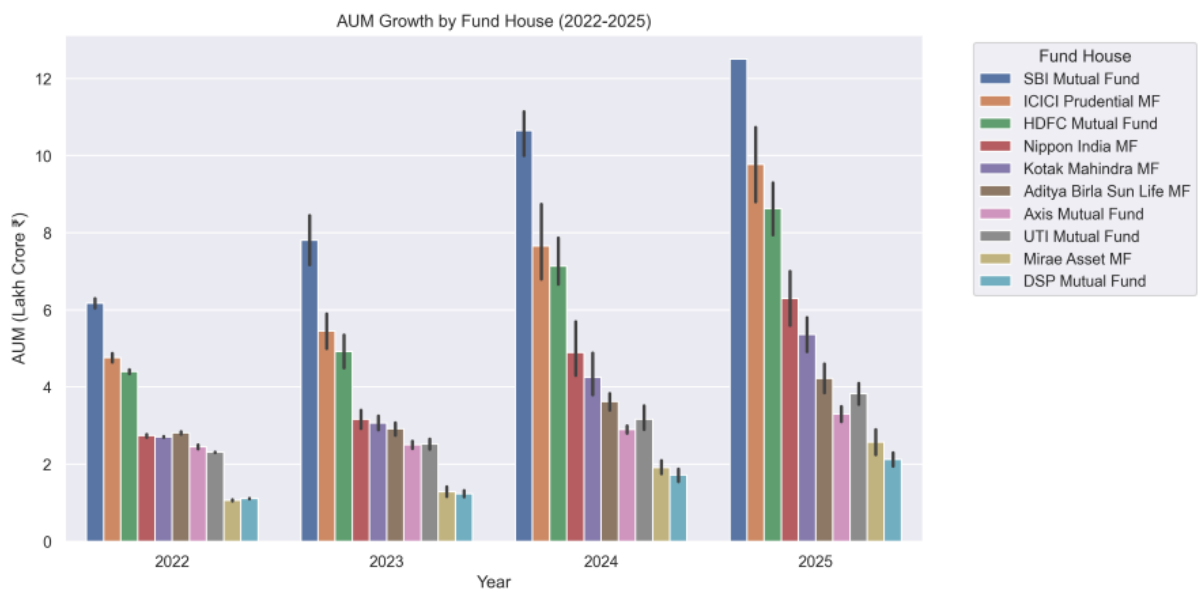
Exploratory Data Analysis

Objectives

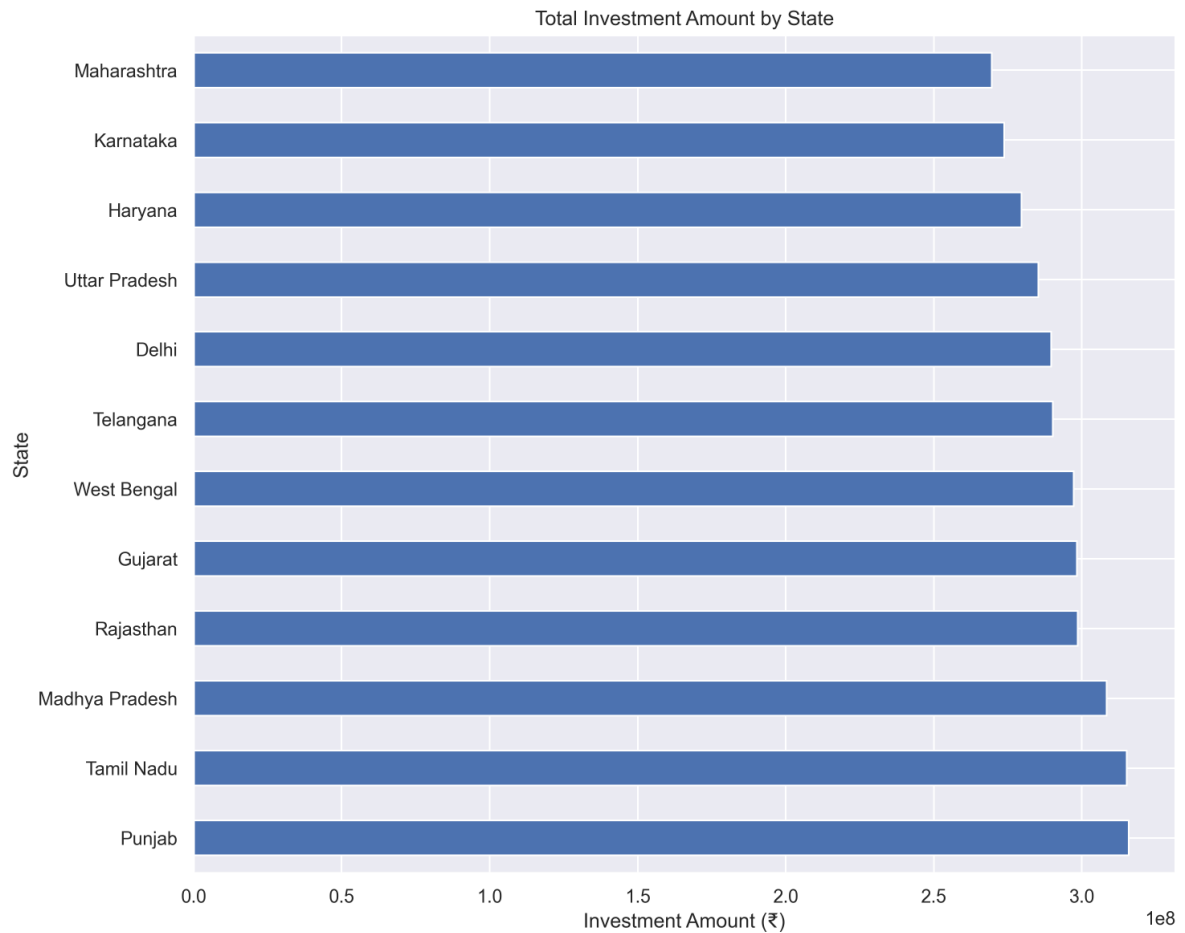
- Understand fund performance trends
- Analyze investor demographics
- Study SIP growth patterns
- Evaluate category inflows
- Examine sector allocations

Key Visualizations

- AUM Growth Trend
- SIP Inflow Trend
- Age Group Distribution
- Gender Distribution
- State Investment Distribution
- Category Heatmap
- NAV Correlation Matrix
- Sector Allocation Analysis



State Investment Distribution



The chart shows steady growth in mutual fund assets under management from 2022–2025, indicating increasing investor participation.

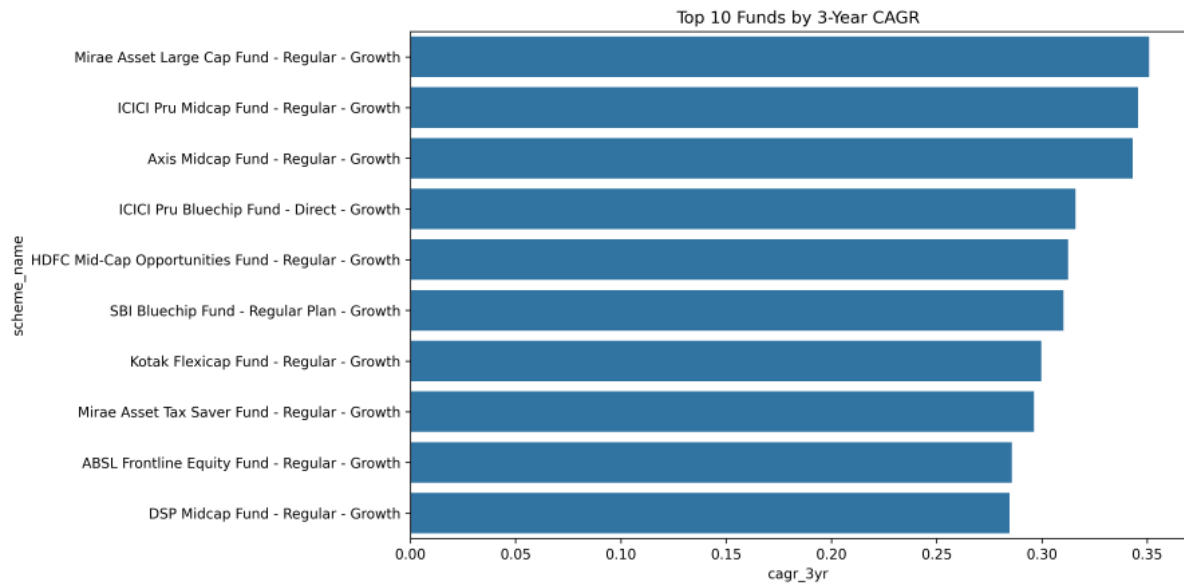
Performance Analytics

Performance metrics were computed for all available mutual fund schemes.

Metrics Calculated

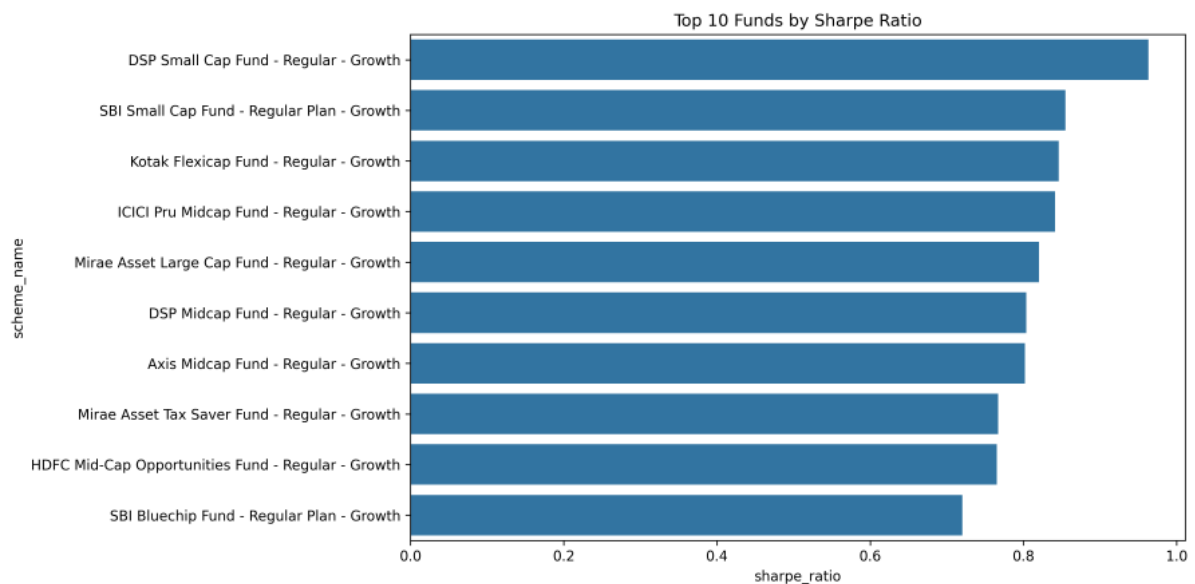
- CAGR
- Alpha
- Beta
- Sharpe Ratio
- Sortino Ratio
- Maximum Drawdown
- Tracking Error

Top 10 Funds by CAGR



This chart ranks the top mutual fund schemes based on Compound Annual Growth Rate (CAGR). Higher CAGR values indicate stronger long-term growth performance and wealth creation potential. The leading funds consistently delivered superior annualized returns over the evaluation period.

Top 10 Funds by Sharpe Ratio

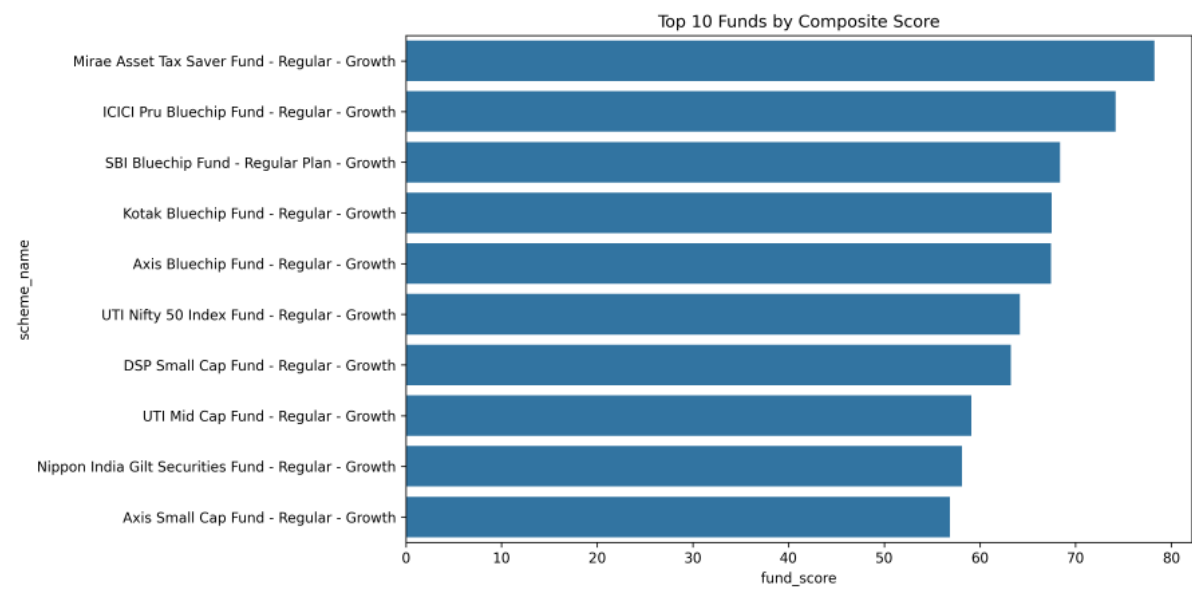


The Sharpe Ratio measures risk-adjusted returns by comparing excess returns to volatility. Funds with higher Sharpe Ratios generated better returns for each unit of risk undertaken. These funds demonstrated strong performance while maintaining relatively efficient risk management.

Top Performing Funds

The analysis identified top-performing funds based on risk-adjusted return metrics and composite fund score rankings.

Fund Scorecard Ranking



The fund scorecard combines multiple performance indicators including returns, Sharpe Ratio, alpha, and risk metrics into a composite ranking. Higher scores identify funds that consistently perform well across multiple evaluation criteria, making them strong candidates for investment consideration.

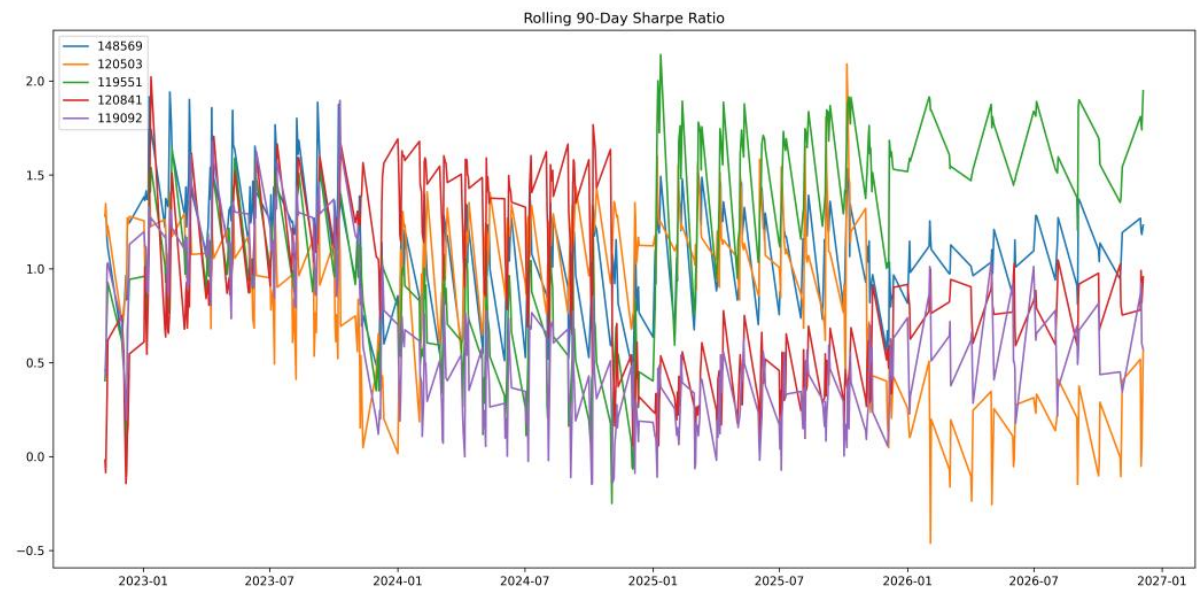
Advanced Analytics & Risk Metrics

Historical Risk Measures

- Value at Risk (VaR 95%)
- Conditional Value at Risk (CVaR)

Rolling Analytics

- Rolling 90-Day Sharpe Ratio



- The Rolling 90-Day Sharpe Ratio tracks changes in risk-adjusted performance over time. Fluctuations in the ratio reflect changing market conditions and fund consistency. Sustained higher values indicate periods of stronger risk-adjusted returns and stable fund performance.

Investor Analytics

- Cohort Analysis
- SIP Continuity Analysis
- Investor Segmentation

Portfolio Analytics

- Sector Concentration Analysis using HHI
- Fund Recommendation Engine based on risk appetite

Dashboard Development

A four-page Power BI dashboard was developed.

Page 1: Industry Overview

- Total AUM
- Total SIP Inflows
- Total Folios
- Total Schemes
- AUM Trend
- AUM by AMC



Total AUM

Sum of ...
39M

Total SIP Inflows

Max of ...
31K

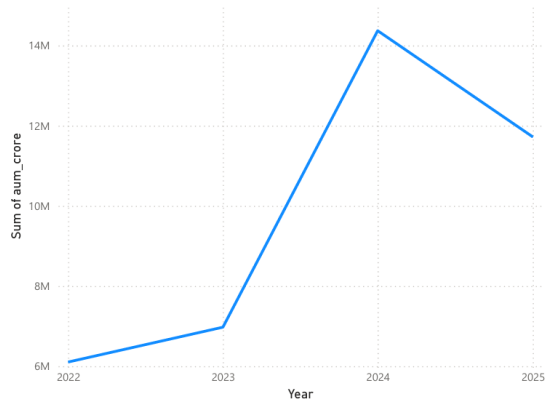
Total Folios

Max of ...
26.12

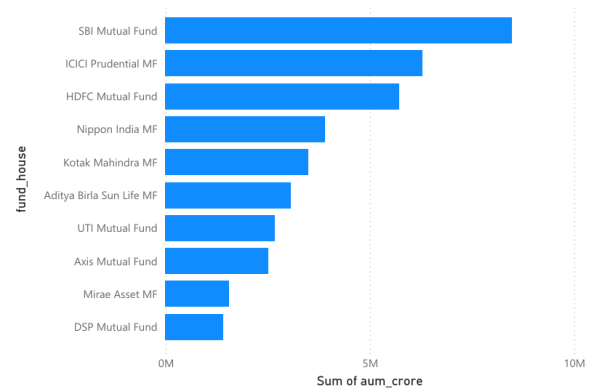
Total Schemes

Count o...
40

Industry AUM Trend (2022–2025)



AUM by AMC



Page 2: Fund Performance

- Risk vs Return Scatter Plot
- Fund Scorecard Ranking
- NAV Trend vs Benchmark
- Interactive Slicers



fund_house

All

category

All

plan

All

risk_grade

All

scheme_name

All

Risk vs Return Analysis

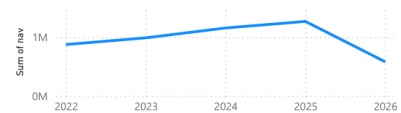
category ELSS Flexi Cap Gilt Index Index/ETF Large & Mid Cap Large Cap Liquid Mid Cap Short Duration Small Cap Value



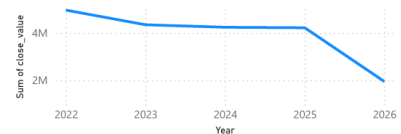
Fund Scorecard Ranking

scheme_name	fund_house	category	fund_score	sharpe_ratio
SBI Small Cap Fund - Direct Plan - Growth	SBI Mutual Fund	Small Cap	29.75	0.9
Kotak Flexicap Fund - Regular - Growth	Kotak Mahindra MF	Flexi Cap	31.00	0.9
Kotak Emerging Equity Fund - Regular - Growth	Kotak Mahindra MF	Mid Cap	32.00	0.9
ABSL Small Cap Fund - Regular - Growth	Aditya Birla Sun Life MF	Small Cap	34.25	0.9
SBI Small Cap Fund - Regular Plan - Growth	SBI Mutual Fund	Small Cap	38.12	0.9
Mirae Asset Large Cap Fund - Regular - Growth	Mirae Asset MF	Large Cap	38.56	1.0
HDFC Short Term Debt Fund - Regular - Growth	HDFC Mutual Fund	Short Duration	39.50	1.8
ICICI Pru Liquid Fund - Regular - Growth	ICICI Prudential MF	Liquid	40.75	7.6
ICICI Pru Midcap Fund - Regular - Growth	ICICI Prudential MF	Mid Cap	41.50	0.9
ICICI Pru Bluechip Fund - Direct - Growth	ICICI Prudential MF	Large Cap	42.88	1.0

Selected Fund NAV Trend



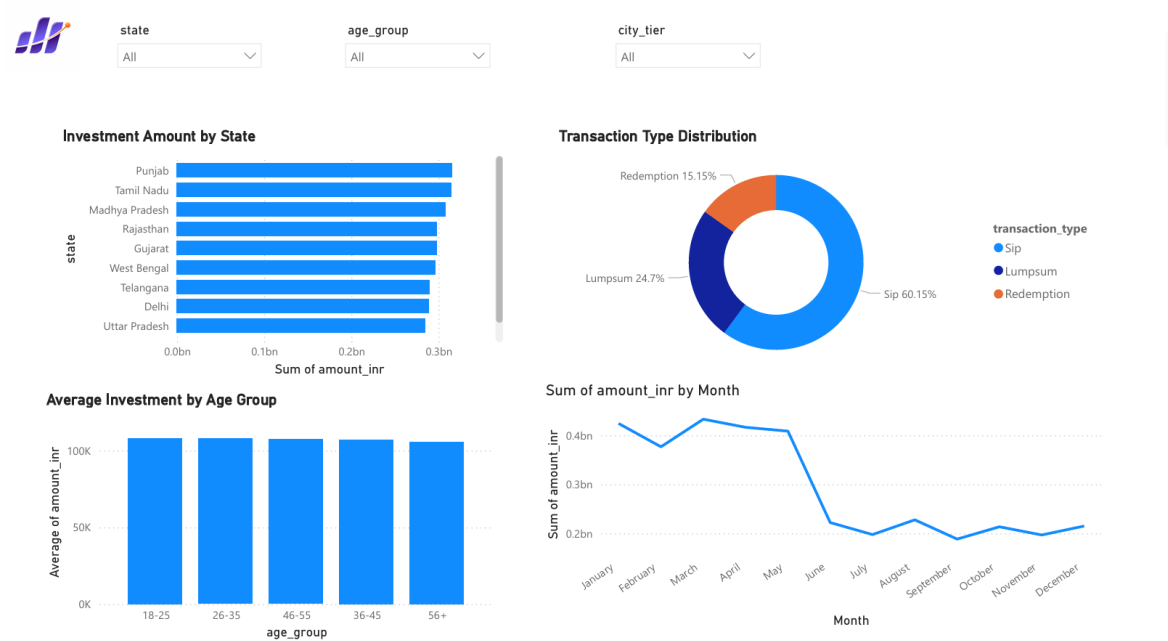
NIFTY100 Benchmark Trend



Page 3: Investor Analytics

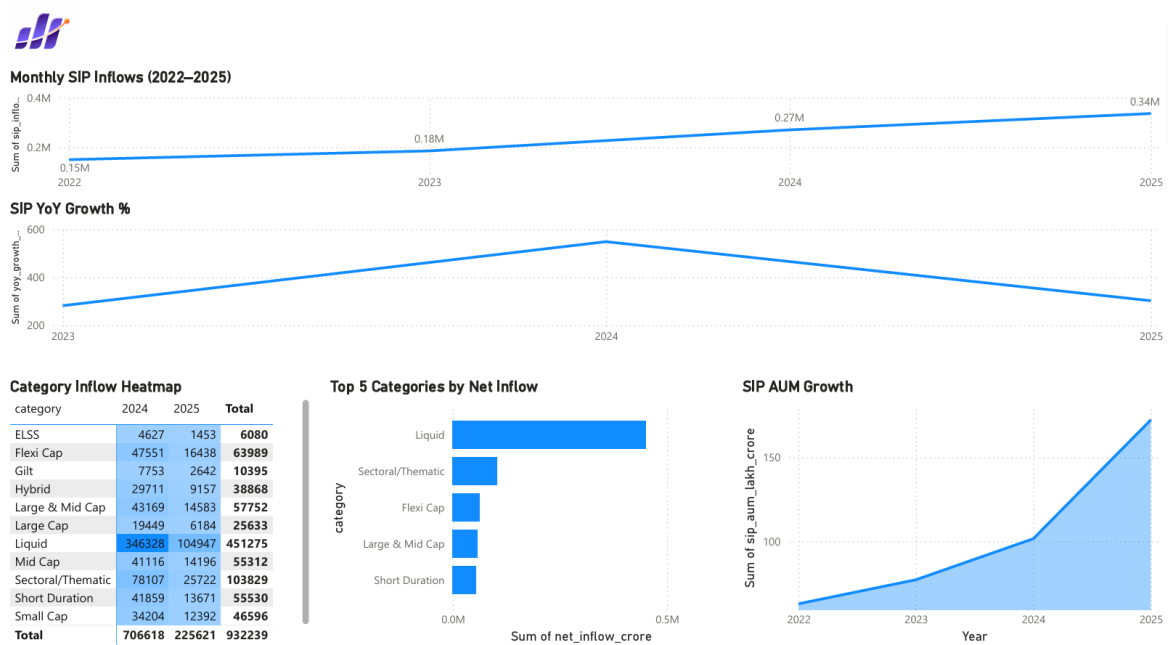
- Investment by State

- Transaction Type Distribution
- Average Investment by Age Group
- Monthly Transaction Trends



Page 4: SIP & Market Trends

- SIP Inflow vs Nifty 50
- Category Inflow Analysis
- Top Categories by Net Inflow



Key Findings

1. SIP inflows demonstrated sustained growth over the analysis period.
2. Equity-oriented schemes generated superior long-term returns.
3. Small-cap funds exhibited higher return potential but increased volatility.
4. Investor participation was highest among working-age demographics.
5. Risk-adjusted performance metrics highlighted a small group of consistently outperforming funds.

Conclusion

The project successfully demonstrates the design and implementation of a complete mutual fund analytics platform. By combining data engineering, financial analytics, advanced risk measurement, and business intelligence dashboards, the solution provides valuable insights for investment analysis and decision-making.